The Influence of Social Consensus Cues on False Recognition of Visually Similar English Words

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Abstract. People sometimes report having seen a word that was never shown, especially when it closely resembles a studied word. We propose an online within-participant experiment that tests whether information about how other people responded changes how often such look-alike words are falsely recognised. Participants study made-up and real words, complete a one-minute filler task, and judge 60 words as old or new, most of them with a simulated report that a high number (8 to 10) or a low number (1 to 3) of 10 earlier participants said old or new, and one in five with no report. The critical items are real words one letter away from a studied made-up word, all taken from a published list (Bowers, Davis, and Hanley, 2005). With 60 participants the study is well powered for the direction of the majority and for large interaction effects only, and we state this limit in advance.

1 Introduction

Memory is reconstructive. People recognise words that never appeared, often with confidence, as in the procedure in which a list of words related to a missing word leads many participants to report that the missing word was on the list (Deese, 1959; Roediger and McDermott, 1995). Meaning is not required for this. Gatti et al. (2025) had participants study lists of meaningless word-like strings and found that new strings were called old more often the more closely they resembled the list, so visual similarity alone can produce false recognition. Judgments are also rarely made alone. Others' judgments shift our own (Sherif, 1935; Asch, 1956), memory reports shift after hearing what others remember (Roediger, Meade, and Bergman, 2001; Gabbert, Memon, and Allan, 2003), and a rating attributed to ten earlier participants, unrelated to the true stimulus, produced large shifts in judgments of pain and effort (Yazdanpanah et al., 2026).

Everyday life places crowd signals next to things we may or may not have seen: messages marked as forwarded many times, download counts, star ratings, and names that look like familiar ones. We bring the two ingredients together. A word that was never studied but looks like one that was is shown with a report of what others said about it, and we measure whether the report changes the tendency to call it old. The aims are to test the effect of the direction and strength of a simulated majority, to describe it relative to a no-cue baseline as creating or suppressing false recognition, and to ask whether it moves the response criterion or the memory evidence.

2 Review of literature

False recognition and similarity. Studying lists of associates of an unpresented word produces false recall and recognition of that word at high rates (Deese, 1959; Roediger and McDermott, 1995). Gatti et al. (2025) adapted the task to pseudowords: 63 native Italian speakers studied 15 lists of 12 pseudowords online, each followed by a two-minute filler task and a test of 12 items. False alarms were .20 for the new item most similar to the list and fell to .09 for the least similar, which the authors attributed to similarity matching at retrieval. Their simulation indicated that about 58 participants gave 80 percent power in their design.

Orthographic neighbours. Bowers, Davis, and Hanley (2005) had participants learn novel words made by changing one internal letter of a familiar six-letter word with no neighbours, for example BANARA from BANANA. This made the familiar words harder to categorise by meaning, with clear interference the next day, which they read as lexical competition. Their appendix lists 40 such pairs and is the source of all our words. We use their items and not their task or interpretation.

Social influence. Sherif (1935) and Asch (1956) showed convergence on others' judgments. Deutsch and Gerard (1955) separated informational influence, where others' answers are treated as evidence, from normative influence, where a person goes along to fit in. Conformity is reliable but varies with setting (Bond and Smith, 1996). In memory, co-witnesses come to report details they did not see (Gabbert et al., 2003), others' errors enter a person's own reports (Roediger et al., 2001), and recognition responses move toward those of others (Schneider and Watkins, 1996).

A minimal consensus cue. In Yazdanpanah et al. (2026), 111 participants saw ten dots described as the mean rating of ten earlier participants before each trial of three tasks (heat pain, others' pain, mental effort). The cue was random with respect to the stimulus. It shifted expectations and experienced ratings, with effect sizes of 0.73 to 0.81 on experience ratings, persisted over three days with some fading, and individual susceptibility was fairly stable. We adopt this cue format.

3 Research gap, questions and hypotheses

We found no study that varies both the direction and the strength of a minimal consensus cue for false recognition of visually similar words with a no-cue baseline in the same session (our searches covered titles and abstracts

only). Work on similarity contains no social cue, and work on the minimal cue concerns ratings and not recognition. It is also unknown whether a majority acts only on the response criterion or is weighed against memory evidence. Methodologically, we use one published word list, a fixed cue table, no experimenter, and a within-person baseline. Our participants are students at an Indian institute who mostly use English as a second language, unlike the base studies. The contribution is a standardised test of when and how a simulated crowd turns a never-studied near-miss word into a false recognition, not a claim of priority.

The research questions are: (1) Do the direction (others say old or new) and strength (high, 8 to 10 of 10, or low, 1 to 3 of 10) of a simulated majority change false recognition of unseen one-letter neighbours of studied pseudowords? (2) With no social information, are these neighbours falsely recognised more than unrelated real words? (3) Relative to no cue, does the majority create false recognition, suppress it, or both? (4) Does it move only the response criterion, or also the memory signal? (5) Do participants respond to the answer shown or to the votes it implies?

If the majority only shifts the criterion, old responses should rise and fall together for all word kinds with unchanged sensitivity. If the report is weighed against memory evidence, the effect should be larger for look-alike words than for clear cases (Deutsch and Gerard, 1955). If participants use the displayed answer, a low report such as "2 of 10 said old" should pull toward old more than "2 of 10 said new"; if they use the implied remainder, the order reverses.

	Hypothesis	Status
H0	Look-alike real words (lures) are called old more often than unrelated real words.	Gatekeeper
H1	Lures are called old more often when others said old than when they said new.	Confirmatory
H2	More agreement pulls further in the stated direction (high-old above low-old, high-new below low-new).	Pre-specified
H3	Lures are called old more often in the high-old condition (a) and less often in the high-new condition (b) than with no cue.	Pre-specified (a), exploratory (b)
H4	The cue shifts the criterion and may change sensitivity; its effect is larger for lures than for unrelated real words.	Exploratory
H5	The low-old minus low-new contrast shows whether the displayed answer or the implied votes drive responses.	Exploratory
H6	The cue effect varies with belief in the other participants, English reading, and trial position.	Exploratory

H1 is the single confirmatory test, and H0 is a gatekeeper: if lures are not called old more often than unrelated words, cue effects on lures are read as effects on errors in general. Strength and direction are not independent here, because the four cue conditions correspond on average to 9, 2, 1, and 8 votes for old. H2 is therefore read as whether the pull grows with the agreement shown, not as a clean effect of strength.

4 Method

Design. One group of participants completes a single online session. Each word at test carries one of five conditions: high-old (for example "9 of 10 other participants said OLD"), low-old ("2 of 10 . . . OLD"), high-new ("9 of 10 . . . NEW"), low-new ("2 of 10 . . . NEW"), or no cue (no line shown). The number shown varies at random within each level (8, 9, or 10 for high; 1, 2, or 3 for low), equally often within each condition, so that the same two numbers do not repeat in an obvious pattern. The 60 test words are of five kinds, which are trial types and not factors: studied pseudowords (15), studied real words (10), lures (15), unrelated real words (10), and unrelated pseudowords (10). Pseudowords and lures fall three times in each condition, the other kinds twice, giving 12 trials per condition. A version number from 0 to 4 rotates which word takes which condition, so every word appears in every condition equally often across participants, and the old and new keys are counterbalanced (ten cells, filled in turn). Test order is random with no more than three words of one kind or four cues of one direction in a row.

Participants. Students of the institute, 18 or older, who read English comfortably and use a laptop or desktop with a keyboard, recruited by convenience through posters and student groups, not through instructors and without course credit. The target is 60 recruited, about 52 usable if about 13 percent are excluded. Recruitment runs for a fixed window, results are not examined to decide when to stop, and the achieved number is reported. The first five or six participants form an internal pilot.

Materials. All words come from the Bowers et al. (2005) appendix (39 usable pairs), each pair given one role by a seeded search that balances the natural and artefact categories, keeps words from different pairs at least three letters apart, and keeps pseudowords resembling Hindi words, a common surname, or slang out of the foil

and practice roles. Lures have a mean letter-sequence similarity of 0.62 to their studied pseudoword and foils at most 0.23 to any studied word. Frequency is not matched, so log frequency rank is a covariate.

Role	Words
Studied pseudoword and its lure (15)	BANARA/BANANA, PIGERN/PIGEON, WALNOT/WALNUT, GARNIC/GARLIC, SPIMER/SPIDER, GANAXY/GALAXY, HOLMET/HELMET, VIODIN/VIOLIN, PUNCIL/PENCIL, NASKIN/NAPKIN, PILTAR/PILLAR, CRAGLE/CRADLE, BALBOO/BAMBOO, TEAPIT/TEAPOT, FABLIC/FABRIC
Studied real word (10)	JACKAL, ANCHOR, FOSSIL, OYSTER, AMAZON, COFFIN, NEEDLE, GUITAR, TENDON, SLEEVE
Unrelated foils, both shown as new (10)	TURKEY/TULKEY, PARCEL/PARGEL, CELERY/CEDERY, ENAMEL/ERAMEL, MEADOW/MEAROW, JERSEY/JERGEY, DIESEL/DIRSEL, KIDNEY/KIDLEY, TOMATO/TORATO, MOSAIC/MOTAIC

Real words are studied and pseudowords appear as foils for a reason. If all studied items were pseudowords and all lures real words, lexical status would reveal old status and suppress false recognition. Here 60 percent of non-word test items were studied against 29 percent of real-word items, a reduced but not removed link, so lexical status is a covariate and the main contrasts compare real words with real words.

Procedure (about 15 minutes). Participants agree to an information page, read short instructions explaining that ten earlier participants did the task, and complete a short practice. They then see the 25 study words (15 pseudowords, 10 real words) one at a time for 5 seconds each, about two minutes, and are asked to remember them exactly. A one-minute filler task follows (a circle or square each second, space bar for circles only), which prevents holding the list in mind and rejecting every lure. At test, each trial shows a 500 ms fixation, then the word with its cue line, if any, until the participant presses F or J for old or new, then a 300 ms blank. Three unannounced checks, each after a cued trial, ask what the others said about the last word and how many of ten said it. Questions follow on what the study was about, how real the others' answers seemed and how much they influenced the participant (seven-point scales), language background, age, gender, programme, and prior participation, and then a debrief.

Measures and exclusions. The primary measure is an old response to a lure. Secondary measures are hits, false alarms to unrelated words, and reaction time, which includes cue reading time and is used only for trial exclusion and exploratory comparisons. Fixed before data collection, participants are excluded for incomplete sessions, fewer than two of three attention checks fully correct, a hit rate not above the false alarm rate, prior participation, or the page being hidden on more than 10 percent of test trials. Trials under 300 ms or over 5000 ms are excluded, the window of Gatti et al. (2025). Participants who doubt the cover story are kept and the analysis is repeated without them.

Analysis. H0 compares lures with unrelated real words in a logit mixed model with participant and item effects and a by-item test, and both must support it. H1 uses lure trials only, as a mixed logistic regression with sum-coded factors, fitting the maximal random effects first and simplifying only if needed (Barr et al., 2013):

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old ~ valence * strength + (1 + valence * strength | participant) + (1 | item)
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H2 and H3 use the same model and within-participant comparisons with no-cue lures, with Holm adjustment over H1, H2, and H3a and 95 percent intervals. H4 uses a probit mixed model on all test trials with old status, cue, and their interaction, where the cue main effect indicates a criterion shift and the interaction a change in sensitivity (DeCarlo, 1998; Macmillan and Creelman, 2005). H5 is the low-old minus low-new contrast with a model on the number of votes for old shown (which varies within each condition). Items are random effects, and claims are limited to similar one-letter neighbours (Clark, 1973).

Sample size. Sixty is the number that can be recruited in the available time. It was not derived from a power analysis and is a resource constraint (Lakens, 2022). It is close to the 63 participants of the study the task is based on, and a within-participant design gives each participant 12 trials per condition. We simulated what it can detect (2000 studies per cell; two fixed cue levels; participant-level approximations of the mixed models; participant intercept SD 0.8 and cue susceptibility SD 0.4 on the logit scale; item SD 0.35 and 0.70; false recognition assumed at .20 for lures and .10 for unrelated words; type I error .04 to .05). Effect sizes are assumptions because no pilot data exist.

Test	Effect (logit)	Power at 52 (item SD .35 / .70)	At 45
H1, direction	0.8	.98 / .97	.96 / .94
H2, interaction	1.0 and 1.4	.70 / .68 and .93 / .91	.65 / .62 and .88 / .89
H3a, high-old against no cue	0.65 and 0.75	.72 / .70 and .84 / .79	.65 / .61 and .79 / .75
H0, lures against real foils	0.8	.94 / .65	.94 / .62

Direction (H1) is well powered; the interaction and the baseline contrasts only for large effects, and H0 depends on item variability. Smaller effects are reported as estimates with intervals, and a non-significant result is never read as no effect. The pilot proceeds unchanged if lure false recognition is at least about 8 percent and above that for unrelated words, foils are above about 5 percent, median completion is about 20 minutes or less, and overt suspicion is below about 30 percent. If lures are at floor, the more similar half of the lure set is used or the filler lengthened.

Ethics and platform. Participants are adults and take part voluntarily, with no anticipated risk beyond mild fatigue. The study involves mild deception, because the answers attributed to others are produced by the researchers. The information page says that some details cannot be given in advance and will be explained at the end, and the debrief explains that the answers were simulated and why, notes that saying old to a look-alike word is a common and normal error, and asks participants not to describe the study to others. No names, emails, or roll numbers are collected. Answers are stored under a random participant code shown at the end, which allows later removal on request, together with basic technical information and a record of how far each session progressed without answers. Quitting leaves no answers. Data are held in an account controlled by the research team under the institute's data guidelines, and recruitment starts only after approval of the institute review board. The task runs in a browser using jsPsych (de Leeuw, 2015), with identical instructions, timing, and layout for all participants, a fixed cue table, a stored seed for each participant, and no experimenter present.

5 Limitations and caveats

- **Sample size.** Sixty recruited, about 52 usable, supports the main effect and large interactions, not small ones, which are reported as estimates with intervals.
- **Strength and direction overlap** (on average 9, 2, 1, and 8 votes for old). They cannot be separated without changing the levels, so the low-old minus low-new contrast and a model on votes are pre-specified.
- **Lexical status** remains weakly linked to old status (60 against 29 percent); it is a covariate and the key contrasts compare real words with real words.
- **No-cue trials sit inside a cued session**, so they are a within-session baseline and not behaviour with no social information at all.
- **Floor effects.** Reports of new push old responses toward zero, so smaller differences in negative conditions are hard to see, and the high-new contrast is exploratory.
- **A cue unrelated to the truth** may be noticed and discounted; belief ratings, a sensitivity analysis, and trial position address this.
- **Items and sample.** Fifteen lures and three trials per cell limit generalisation across words; the sample is convenience-based, mostly second-language speakers, tested online without supervision; the word list comes from a lexical-competition study and is cited for its items only; repeat participation from another browser cannot be ruled out.

6 Implications and extensions

If the hypotheses are supported, a minimal and unrelated majority can turn a never-studied near-miss word into a false recognition, with the effect growing with the agreement shown. Next steps would separate strength from direction with split votes or a third answer, vary who the crowd is (own group, outsiders, experts), delay the test by a day (Bowers et al., 2005), add confidence ratings to separate criterion from memory evidence, and extend to other languages and real-world material. If they are not supported, the intervals show which effect sizes are excluded; low belief ratings would call for a more credible cover story, and near-zero false recognition without a cue for more similar lures or a longer delay. In every case the materials and data would be shared openly.

References

Asch, S. E. (1956). Studies of independence and conformity: I. A minority of one against a unanimous majority. *Psychological Monographs, 70*(9, Whole No. 416).

Barr, D. J., Levy, R., Scheepers, C., and Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language, 68*, 255-278.

Bond, R., and Smith, P. B. (1996). Culture and conformity: A meta-analysis of studies using Asch's line judgment task. *Psychological Bulletin, 119*, 111-137.

Bowers, J. S., Davis, C. J., and Hanley, D. A. (2005). Interfering neighbours: The impact of novel word learning on the identification of visually similar words. *Cognition, 97*, B45-B54. `https://doi.org/10.1016/j.cognition.2005.02.002`

Clark, H. H. (1973). The language-as-fixed-effect fallacy. *Journal of Verbal Learning and Verbal Behavior, 12*, 335-359.

de Leeuw, J. R. (2015). jsPsych: A JavaScript library for creating behavioral experiments in a Web browser. *Behavior Research Methods, 47*, 1-12.

DeCarlo, L. T. (1998). Signal detection theory and generalized linear models. *Psychological Methods, 3*, 186-205.

Deese, J. (1959). On the prediction of occurrence of particular verbal intrusions in immediate recall. *Journal of Experimental Psychology, 58*, 17-22.

Deutsch, M., and Gerard, H. B. (1955). A study of normative and informational social influences upon individual judgment. *Journal of Abnormal and Social Psychology, 51*, 629-636.

Gabbert, F., Memon, A., and Allan, K. (2003). Memory conformity: Can eyewitnesses influence each other's memories for an event? *Applied Cognitive Psychology, 17*, 533-543.

Gatti, D., Petilli, M. A., Marchetti, M., Vecchi, T., Mazzoni, G., Rinaldi, L., and Marelli, M. (2025). False memories from nowhere: Humans falsely recognize words that are not attested in their vocabulary. *Psychonomic Bulletin and Review, 32*, 1922-1931. https://doi.org/10.3758/s13423-025-02677-7

Lakens, D. (2022). Sample size justification. *Collabra: Psychology, 8*(1), 33267.

Macmillan, N. A., and Creelman, C. D. (2005). *Detection theory: A user's guide* (2nd ed.). Lawrence Erlbaum.

Roediger, H. L., and McDermott, K. B. (1995). Creating false memories: Remembering words not presented in lists. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 21*, 803-814.

Roediger, H. L., Meade, M. L., and Bergman, E. T. (2001). Social contagion of memory. *Psychonomic Bulletin and Review, 8*, 365-371.

Schneider, D. M., and Watkins, M. J. (1996). Response conformity in recognition testing. *Psychonomic Bulletin and Review, 3*, 481-485.

Sherif, M. (1935). A study of some social factors in perception. *Archives of Psychology, 27*(187), 1-60.

Yazdanpanah, A., Jung, H., Soltani, A., and Wager, T. D. (2026). Social information creates self-fulfilling prophecies in judgments of pain, vicarious pain, and cognitive effort. *Proceedings of the National Academy of Sciences, 123*(7), e2513856123. https://doi.org/10.1073/pnas.2513856123